Detector-Plane Causality and Imprint Dynamics:

A Unified Measurement Framework for Quantum Coherence

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Abstract

We present a unified framework for quantum measurement that integrates particle-based field excitation, detector-plane causality, and coherence-resolved imaging into a single operational model. The framework combines three complementary components: (1) the Imprint Hypothesis, which treats quantum particles as primary field excitations and wave phenomena as contextual imprint propagation; (2) Detector-Plane Causality, formalized through the Quantum Measurement Stack (QMS) and QMCTB-01 benchmark, demonstrating that interference visibility is determined by detector-plane coherence preservation; and (3) Detector-Plane Imaging (DPI), which shows that spatial coherence structures evolve continuously at the detector plane prior to discretization.

Using nonlinear scalar-field simulations and detector-aware modeling, we demonstrate that interference patterns are encoded in pre-measurement coherence, decoherence is a continuous dynamical process, and measurement outcomes are causally determined by detector architecture. This framework reconciles quantum optics, statistical optics, and detector engineering within a QFT-consistent, experimentally testable model.

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1. Introduction

Quantum measurement has historically been interpreted through probabilistic accumulation, observer-dependent collapse, and wave–particle duality. However, modern experiments reveal persistent ambiguities:

- Interference appears to “build up” over time
- Detection outcomes depend strongly on instrumentation
- Coherence and decoherence lack a unified physical description

This work resolves these issues by introducing a measurement-plane framework grounded in three principles:

1. Particle primacy — quantum objects are field excitations
2. Wave contextuality — wave behavior arises from field propagation
3. Detector causality — measurement outcomes are determined locally by detector systems

We unify these into a single operational structure spanning field dynamics, measurement physics, and detector-plane imaging.

2. The Imprint Hypothesis: Particle Primacy and Wave Contextuality

The Imprint Hypothesis establishes the ontological foundation:

- Quantum fields are the fundamental substrate
- Particles are discrete excitations created via field operators
- Wave-like behavior emerges from field propagation (Green's functions)

Key Principle

Wave phenomena (interference, diffraction, tunneling) are contextual imprints generated by particle–field interactions.

Measurement collapses the contextual imprint, fixing observable outcomes without negating particle existence.

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3. Detector-Plane Causality: Measurement as a Local Process

Measurement is formalized through the Quantum Measurement Stack (QMS):

1. Field excitation and imprint formation
2. Detector-plane interaction
3. Data acquisition (DAQ)
4. Signal processing
5. Analysis

The detector plane defines the effective measurement boundary where information is irreversibly transformed.

QMCTB-01 Benchmark

Controlled double-slit simulations demonstrate:

- Identical propagation and photon statistics
- Detector-dependent outcomes

Results:

- Intensity-only detection → no interference
- Correlation-preserving detection → high-visibility interference
- No dependence on integration time

Core Conclusion

Interference visibility is determined by detector-plane coherence preservation, establishing detector-plane causality as the mechanism of measurement.

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4. Detector-Plane Imaging (DPI): Coherence Before Discretization

DPI treats the detector surface as an imaging geometry that records evolving spatial coherence prior to discretization.

The detector records:

$I(x, y, t) = \varphi^2(x, y, z_0, t)$

This represents a continuous imprint field rather than discrete detection events.

Observations

Coherent regimes: - Concentric coherence rings
- Long-range spatial correlations
- Narrow spectral peaks

Decoherent regimes: - Fragmentation of coherence structure
- Spectral broadening
- Stochastic detector-plane textures

Key Insight

Interference patterns reflect real spatial structure present at the detector plane prior to discretization.

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5. Coherence and Decoherence Dynamics

Coherence corresponds to sustained phase alignment and extended spatial correlations.

Decoherence arises from nonlinear mode coupling and energy redistribution, producing:

- Correlation-length decay $\xi(t)$
- Fourier spectral broadening
- Loss of structured spatial order

Decoherence is a continuous dynamical process rather than an abrupt transition.

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6. Double-Slit Interference Reinterpreted

Using DPI-derived fields as illumination:

Coherent illumination: - High-visibility fringes
- Deep minima
- Structured diffraction envelopes

Decoherent illumination: - Reduced fringe contrast
- Lifted minima
- Suppressed interference structure

The double slit reveals coherence already present in the field rather than generating it.

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7. Unified Measurement Model

1. Particle excitation as a field excitation
2. Imprint formation through propagation
3. Detector-plane sampling of the imprint
4. Detector-dependent filtering of correlations
5. Discrete observed outcome

Measurement is a causal filtering process applied to a pre-existing coherence structure.

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8. Experimental Predictions

1. Detector dependence of interference visibility
2. Exposure-dependent coherence length $\xi(t)$
3. Local suppression of interference via detector-plane decoherence
4. Pre-discretization spatial coherence structure

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9. Applications

- Quantum computing: detector-aware error correction
- Quantum sensing: calibrated decoherence and noise characterization
- Imaging systems: coherence-aware detector design
- Semiconductor physics: tunneling as imprint-mediated transport

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10. Conceptual Implications

- Resolves wave–particle duality via particle primacy

• Defines collapse as a detector-plane process
• Reinterprets interference buildup as statistical convergence
• Establishes measurement as an instrumentation problem

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11. Conclusion

This unified framework establishes detector-plane causality and imprint dynamics as the operational basis of quantum measurement. By integrating particle excitation, field coherence, and detector physics, it provides a QFT-consistent and experimentally testable model that transforms quantum measurement from an interpretive problem into a controllable physical process.

References and Resources

Imprint Hypothesis: https://doi.org/10.5281/zenodo.18383169
QMCTB-01 Benchmark: https://doi.org/10.5281/zenodo.18075090
Code: https://github.com/srikarr20/quantum-measurement-stack